Problem 6

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Problem 1 *A piecewise defined function f is given by*

$$f(x) = \begin{cases} \dfrac{2x-3}{x-2} & \textit{if} \quad x < 2 \\ \dfrac{x^2-5x+6}{x^2-4} & \textit{if} \quad x > 2 \end{cases}$$

Determine the form of the limit, then find the limit.

(a) $\lim x \to 2^+ f(x)$

Explanation. Since $\lim x \to 2^+ f(x) = \lim x \to 2^+ \dfrac{x^2-5x+6}{x^2-4}$, the form of the limit is $\dfrac{0}{0}$.

$\lim x \to 2^+ f(x) = \lim x \to 2^+ \dfrac{x^2-5x+6}{x^2-4} = \lim x \to 2^+ \dfrac{(x-3)(x-2)}{(x-2)(x+2)} = \lim x \to 2^+ \dfrac{x-3}{x+2} = -\dfrac{1}{4}$

(b) $\lim x \to 2^- f(x)$

Explanation. Since $\lim x \to 2^- f(x) = \lim x \to 2^- \dfrac{2x-3}{x-2}$, the form of the limit is $\dfrac{\#}{0}$.

$\lim x \to 2^- f(x) = \lim x \to 2^- \dfrac{2x-3}{x-2} = -\infty,$

since the numerator positive, the denominator negative and goes to 0.

(c) $\lim x \to 2 f(x)$

Explanation. The limit does not exist, because $\lim x \to 2^+ f(x) \neq \lim x \to 2^- f(x)$.